

Dimitris Mouris

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Applied cryptography researcher motivated by the impact of cryptography in contemporary society and on humanity.

Research & Work Experience

Nillion (Nilogy Inc.)

HEAD OF CRYPTOGRAPHY

Remote

June 2025 – Now

SENIOR CRYPTOGRAPHY RESEARCHER (R&D)

January 2024 – June 2025

I lead cryptography R&D, advancing zkTLS, privacy-preserving LLM protocols with MPC and FHE, and large-scale aggregation primitives (e.g., Mastic, PLASMA). I also support the MPC-based database and TEE-based Confidential VM efforts, alongside papers, talks, demos, and partnerships.

📄 zkTLS 1, 2, 3 📄 MPC-TEEs, 📄 Wave, 📄 Ripple, 📄 Curl, 📄 FHE Summit

University of Delaware

RESEARCH & TEACHING ASSISTANT

Newark, DE, USA

February 2019 – December 2023

Applied Cryptography • Secure Multiparty Computation • Homomorphic Encryption • Zero-knowledge Proofs

Meta Inc.

PART-TIME STUDENT RESEARCHER (REMOTE)

Bellevue, WA, USA & Remote

September 2022 – December 2022

RESEARCH ENGINEER INTERN

June 2022 – August 2022

Development of privacy-enhancing solutions that solve real-world problems such as private attribution and privacy-preserving personalization. Research and implementation of *Delegated Private Matching for Compute*.

📘 Blogpost, 🗣️ Private-ID

Amazon Web Services (AWS)

SOFTWARE DEVELOPMENT ENGINEER (SDE) INTERN

Palo Alto, CA, USA

June 2021 – August 2021

Development of Role-Based Access Control (RBAC) for Amazon Redshift to manage permissions and security privileges.

Athena Research & Innovation Center

RESEARCH ASSISTANT & SOFTWARE ENGINEER

Athens, Greece

September 2017 – December 2018

My Health My Data (MHMD) project: Design and implementation of an end-to-end framework for privacy-preserving medical data analytics (i.e., histograms and decision trees) using secure multi-party computation (MPC).

🗣️ smpc-analytics

Education

Department of Electrical & Computer Engineering, University of Delaware

DOCTOR OF PHILOSOPHY (PHD) IN ELECTRICAL AND COMPUTER ENGINEERING

Newark, DE, USA

February 2019 – December 2023

- Research Topic: *Private and Verifiable Computation*
- My research focuses on privacy-preserving technologies and on building efficient solutions that respect data privacy using cryptographic techniques such as *zero-knowledge proofs*, *homomorphic encryption*, and *secure multiparty computation*.

Advisor: Nektarios G. Tsoutsos

Department of Informatics & Telecommunications, University of Athens

MASTER OF SCIENCE (MSc) IN COMPUTER SCIENCE

Athens, Greece

October 2016 – September 2018

The *Computer Systems (Software and Hardware)* specialization covers advanced topics in areas such as computer security & architecture, operating & distributed systems, programming languages, algorithms & data structures.

Department of Informatics & Telecommunications, University of Athens

BACHELOR OF SCIENCE (BSc) IN COMPUTER SCIENCE

Athens, Greece

October 2012 – September 2016

Publications (Selected)

JOURNAL ARTICLES

- C. Gouert, **D. Mouris**, and N. Tsoutsos. “**HELM: Navigating Homomorphic Encryption through Gates and Lookup Tables**” in *IEEE Trans. on Information Forensics and Security (TIFS)*, 📄 2023/1382, 🗣️ 2025
- J. Gao, Y. Palanikumar, **D. Mouris**, D. Nguyen, and N. Trieu. “**SecurED: Secure Multiparty Edit Distance for Genomic Sequences**” in *Proc. Priv. Enhancing Technol. (PoPETs)*, 📄 vol. 2025, no. 2 2025
- **D. Mouris**, C. Patton, H. Davis, P. Sarkar, and N. Tsoutsos. “**Mastic: Private Weighted Heavy-Hitters and Attribute-Based Metrics**” in *Proc. Priv. Enhancing Technol. (PoPETs)*, 📄 vol. 2025, no. 1, 📺 IETF - CFRG, 🗣️ 2025
- C. Gouert, **D. Mouris**, and N. Tsoutsos. “**Juliet: A Configurable Processor for Computing on Encrypted Data**” in *IEEE Trans. on Computers*, 📄 vol. 73, no. 9, 🗣️ 2024
- **D. Mouris**, P. Sarkar, and N. Tsoutsos. “**PLASMA: Private, Lightweight Aggregated Statistics against Malicious Adversaries**” in *Proc. Priv. Enhancing Technol. (PoPETs)*, 📄 vol. 2024, no. 3, 📺 IETF - CFRG, 🗣️ 2024

- D. Mouris, D. Masny, N. Trieu, S. Sengupta, P. Buddhavarapu, and B. Case. “**Delegated Private Matching for Compute**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2024, no. 2,  *Blogpost*,  2024
- C. Gouert, D. Mouris, and N. Tsoutsos. “**SoK: New Insights into Fully Homomorphic Encryption Libraries via Standardized Benchmarks**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2023, no. 3, pp. 154–172,   2023
- D. Mouris, C. Gouert, and N. Tsoutsos. “**Privacy-Preserving IP Verification**” in *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*,  vol. 41, no. 7, pp. 2010–2023 2021
- D. Mouris and N. Tsoutsos. “**Zilch: A Framework for Deploying Transparent Zero-Knowledge Proofs**” in *IEEE Trans. on Information Forensics and Security (TIFS)*,  vol. 16, pp. 3269–3284,  2021

CONFERENCE PAPERS

- M. Ugurbil, D. Mouris, M.B. Santos, J. Cabrero-Holgueras, M. de Vega, and S. Sengupta. “**Fission: Distributed Privacy-Preserving Large Language Model Inference**” in *2025 IEEE CNS*,   2025
- C. Gouert, M. Ugurbil, D. Mouris, M. de Vega, and N. Tsoutsos. “**Ripple: Accelerating Programmable Bootstraps for FHE with Wavelet Approximations**” in *27th Information Security Conference (ISC)*,   *FHE Summit*,  2024
- M.B. Santos, D. Mouris, M. Ugurbil, S. Jarecki, J. Reis, S. Sengupta, and M. de Vega. “**Curl: Private LLMs through Wavelet-Encoded Look-Up Tables**” in *CAMLIS 2025*,   2024
- D. Mouris, C. Gouert, and N. Tsoutsos. “**MPloC: Privacy-Preserving IP Verification using Logic Locking and Secure Multiparty Computation**” in *IEEE 29th Int. Symp. on On-Line Testing and Robust System Design (IOLTS)*,   2023
- D. Mouris, C. Gouert, and N. Tsoutsos. “**zk-Sherlock: Exposing Hardware Trojans in Zero-Knowledge**” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*,  pp. 170-175 2022
- D. Mouris and N. Tsoutsos. “**Pythia: Intellectual Property Verification in Zero-Knowledge**” in *ACM/IEEE 57th Design Automation Conference (DAC)*,  pp. 1–6 2020

THESES

- D. Mouris. “**Private And Verifiable Computation**” in *Ph.D. Thesis*, Advisor: N.G Tsoutsos,  2024
- A. Giannopoulos and D. Mouris. “**Privacy preserving medical data analytics using secure multi party computation. An end-to-end use case**” in *M.Sc. Thesis*, Supervisors: Y. Ioannidis and M. Garofalakis,   2018

INTERNET DRAFTS (IETF)

- H. Davis, D. Mouris, C. Patton, N.G. Tsoutsos. “**The Mastic Verifiable Distributed Aggregation Function (VDAF)**” in *Internet Research Task Force (IRTF)*, Crypto Forum Research Group (CFRG),    2024 – Now

Technical Skills

Programming Paradigms	Procedural  , Object Oriented  , Logic  , Functional 		
Programming Languages	Rust  , C  , C++  , Java  , Python  , Go  , Haskell 		
Parallel Programming	POSIX processes & threads  , MPI  , Open MP 		
Assembly Languages	x86/x64  , MIPS 		
Database Systems	SQL  , MySQL 		
Version Control	Git  , Mercurial 		
	Languages	Greek (<i>native</i>), English (<i>fluent</i>)	

Teaching Assistant

U. of Delaware	Reverse Eng. & Pen. Testing (Fall '21, '22), Microprocessor Sys. (Fall '20), Applied Cryptography (Spr. '20, '21), Secure Soft. Design (Spr. '20, '21), Embedded Systems Sec. (Fall '19 – '21)
U. of Athens	Sys. Prog. (Spr. '17), Logic Prog. (Spr. '17), Intro to Prog. (Fall '14 – '17), Operating Systems (Fall '16)

Honors & Awards

2023	Conference Stipend , Travel stipend for Privacy Enhancing Technologies Symposium (PETS)	Lausanne
2021	Student Conference Grant , Registration for ACM Symp. on Computer and Com. Security (CCS)	Virtual
2021	1st for outstanding research presentation , IEEE Reliable and Resilient Digital Manufacturing	Virtual
2020	Scholarship , Outstanding Academic Performance Scholarship, from the Gerondelis Foundation	Grant \$5,000
2020	DAC Young Fellow Program , 57 th Design Automation Conference, July 20-24	Virtual
2016	Scholarship , Programming Languages Mentoring Workshop (PLMW) at SPLASH 2016, 10/30-11/4	Amsterdam

Community Service

- **Global Co-lead** in BlueHens Capture The Flag (CTF) Competition - UDCTF. 2021, 2022
- **Global Co-lead** in CSAW Cybersecurity Games & Conference – Embedded Security Challenge 2020 – 2023, 2025

Work Authorization

US Permanent Resident via National Interest Waiver based on merit